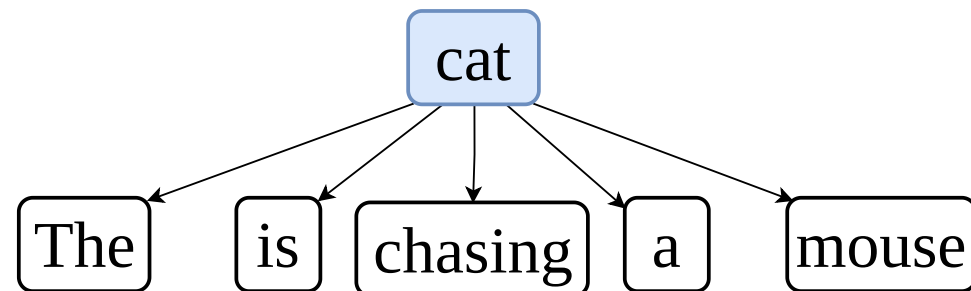


Linguistic Relations

(A) Syntagmatic relation
what can combine?

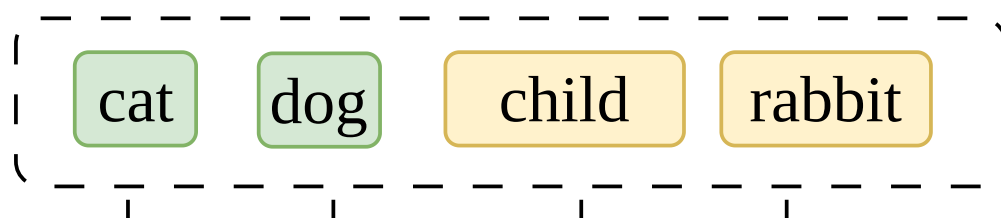
The cat is chasing a mouse



Structural compatibility

(B) Paradigmatic relation
What can substitute?

The [MASK] is chasing a mouse



Same structural slot Different lexical choices

Substitution class

Relational Implementation

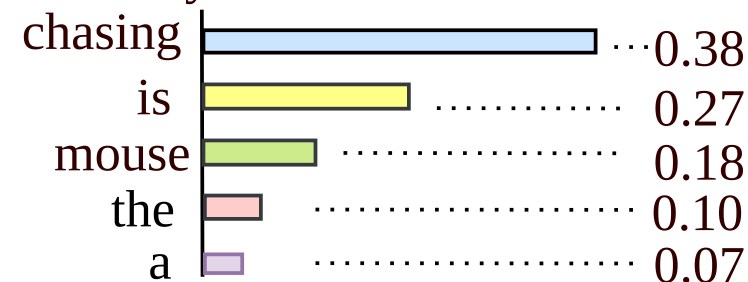
Syntagmatic Module

Attention-guided structural context

Learns structural context compatibility

The [MASK] is chasing a mouse

Last-layer attention from [MASK]



Top-k
context
tokens

Weighted embedding
aggregation



Syn target embedding



Masked hidden state



Syn target embedding

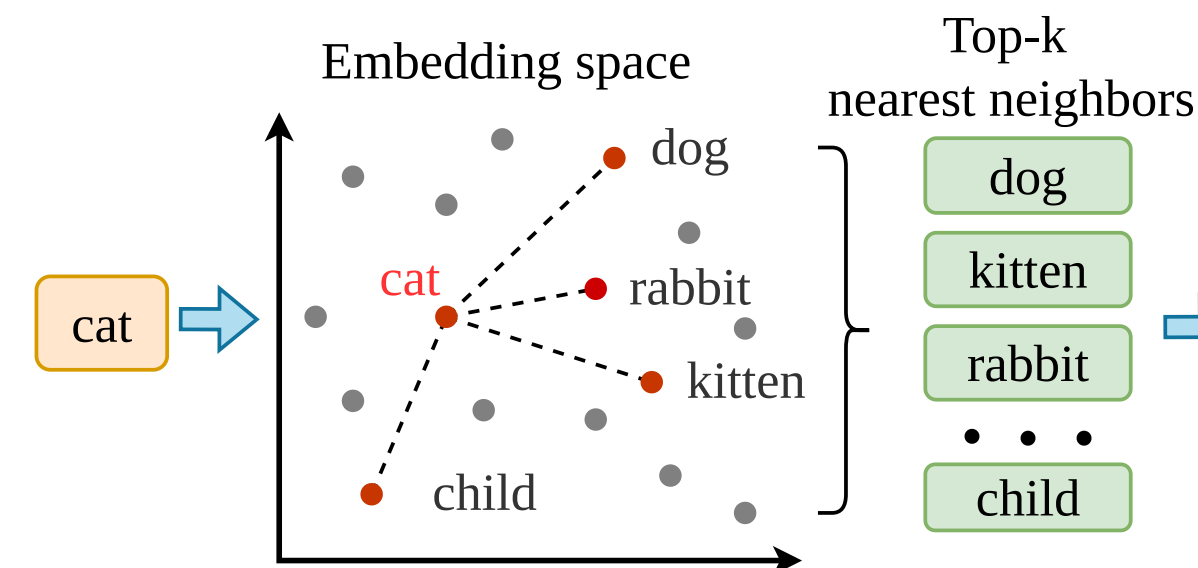


Structural alignment

Paradigmatic Module

Embedding-neighbor substitution

Learns substitution compatibility



Mean neighbor
embedding



Para target embedding



Masked hidden state



Para target embedding



Substitution alignment

Combined

Masked hidden state h_i



Syn target embedding



Para target embedding

